

# Delivering Degrees

## Educational Spillovers from Amazon's Logistics Expansion

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### Abstract

This paper estimates the causal effect of Amazon fulfillment center entry on local educational attainment by exploiting staggered facility openings across U.S. commuting zones between 2010 and 2023. Using a difference-in-differences framework, I find that Amazon's entry increases bachelor's degree attainment by 0.58 percentage points and master's degree attainment by 0.41 percentage points, with no evidence of the increases in high school dropout rates commonly associated with traditional resource booms. Amazon's expansion also raises four-year public college enrollment by 0.52 percentage points. The findings suggest two complementary mechanisms. First, higher household income relaxes financial constraints on educational investment. Second, Amazon's presence generates spillovers to information-intensive industries and high-skill occupations, increasing the returns to postsecondary education. The effects are concentrated among men and White students, indicating heterogeneous responses across demographic groups.

*Keywords:* Amazon; e-commerce; educational attainment; college enrollment; human capital; local labor markets; difference-in-differences

*JEL codes:* I24; I25; J24; L81; R23

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# 1 Introduction

Over the past fifteen years, Amazon has undertaken one of the most dramatic expansions of private logistics infrastructure in American history. Supported by billions of dollars in state and local subsidies ([Good Jobs First, 2022](#)), the company has constructed a sprawling network of fulfillment centers, delivery stations, and air hubs that now spans nearly every major metropolitan area in the United States. This expansion has generated substantial economic activity: the facilities themselves create thousands of relatively well-paid, low-skill jobs with flexible schedules, while also supporting a growing ecosystem of white-collar employment in management, logistics coordination, and technology development ([Jones and Zipperer, 2018](#)). Yet despite the scale of this transformation—and the magnitude of public investment used to attract these facilities—fundamental questions about its long-term impact on local welfare remain unresolved ([Bartik, 2020](#)).

The existing research on Amazon’s local economic effects remains limited. Recent evidence documents that Amazon facility openings increase county-level employment-to-population ratios, reduce poverty rates, and raise median household incomes ([Pathania and Netessine, 2025](#)). Working papers have analyzed employment and wage effects alongside job quality characteristics at fulfillment centers ([Cunningham, 2024](#); [Jones and Zipperer, 2018](#)). Looking at the early expansion of Walmart—a related case of large retail infrastructure growth—studies find that store openings reduced local retail employment by approximately 150 workers per store ([Neumark et al., 2008](#)). What remains absent from this literature is systematic evidence on whether these economic gains translate into human capital formation, a critical determinant of long-run prosperity. Educational attainment and college enrollment are among the most robust predictors of individual economic mobility and regional economic growth. Understanding whether communities that attract large e-commerce distribution facilities experience gains or losses in these measures is therefore essential for evaluating the full social return to the substantial public subsidies these firms command.

This paper addresses the gap by investigating how the arrival of Amazon’s distribution infrastructure affected the educational trajectory of the local population. The theoretical prediction is ambiguous. On one hand, the creation of numerous moderately well-paid jobs requiring minimal formal education could increase the opportunity cost of continued schooling. Young people at the margin of completing high school or enrolling in college might rationally choose immediate employment over further investment in education, particularly if local labor market conditions suddenly improve. This substitution channel echoes findings from an established literature documenting how economic booms—whether driven by natural resource extraction ([Black et al., 2005](#); [Cascio and Narayan, 2022](#)), manufacturing revivals ([Atkin, 2016](#)), or housing market expansions ([Charles et al., 2018](#))—often depress educational attainment by pulling young workers out of school and into the labor force.

On the other hand, Amazon’s entry differs fundamentally from these earlier shocks. The company operates as a technology-intensive logistics firm, not merely a warehouse operator. Its arrival may signal and catalyze broader economic upgrading: attracting complementary high-skill employers, generating productivity spillovers to incumbent firms, and expanding the local demand for educated workers. Moreover, if household incomes rise sufficiently, families may face relaxed financial constraints that enable continued educational investment (Lochner and Monge-Naranjo, 2011; Lovenheim, 2011). Higher local wages may also allow students to fund education through part-time work without delaying enrollment (Johnson, 2013; Scott-Clayton, 2012). These countervailing forces constitute an “upgrading effect,” which could offset or even outweigh the substitution pressures documented in prior work. Which channel dominates is ultimately an empirical question.

This paper estimates the causal effect of Amazon’s fulfillment center expansion on local human capital formation using a novel integration of facility-level data from MWPVL International (MWPVL International, 2025) with educational outcomes from the American Community Survey (ACS), enrollment records from the Integrated Postsecondary Education Data System (IPEDS), and labor market indicators from Lightcast, spanning 170 U.S. commuting zones between 2010 and 2023. The empirical strategy exploits the staggered rollout of Amazon’s distribution network across these commuting zones, using variation in the timing of a fulfillment center’s first entry to identify causal effects in event-study and difference-in-differences (DiD) models. To address well-documented concerns regarding bias in two-way fixed effects estimators arising from staggered treatment timing and heterogeneous treatment effects (Goodman-Bacon, 2021; De Chaisemartin and d’Haultfœuille, 2020; Borusyak et al., 2021; Sun and Abraham, 2021; Athey and Imbens, 2022; Roth et al., 2023), I employ the Callaway and Sant’Anna (2021) estimator, which constructs valid counterfactuals by comparing newly treated commuting zones to those not yet treated at each point in time. Identification rests on the assumption that treated and not-yet-treated commuting zones would have followed parallel trends in educational outcomes absent Amazon’s entry. This assumption receives strong support from both formal balance tests on pre-treatment first-differences (Marcus and Sant’Anna, 2021) and visual evidence of stable, statistically insignificant pre-treatment trends in the event-study plots.

My results show a significant upward shift in the educational distribution following Amazon’s entry. I estimate statistically significant increases in the share of the adult population holding bachelor’s degrees (0.58 percentage points) and master’s degrees (0.41 percentage points). I also detect a modest, marginally significant increase in the share completing high school. The effect on some college attainment is positive but imprecise. Importantly, I find no evidence of rising dropout rates among teenagers—a sharp departure from the substitution effects documented in earlier boom studies. On the enrollment margin, four-year public college participation increases significantly, with effects concentrated among men and White students. Community college enrollment shows posi-

tive but statistically insignificant changes.

Mechanism analysis points to two primary channels driving these results. First, an income channel: Amazon’s entry raises median household income by nearly \$1,000 and reduces the local poverty rate, relaxing the financial constraints that prevent many families from investing in postsecondary education (Lochner and Monge-Naranjo, 2011; Lovenheim, 2011). Second, a demand-side spillover channel: fulfillment center entry reshapes the broader local economy, expanding establishments in the information sector, shifting local employment toward abstract cognitive tasks, and raising the share of technology-related job postings. These structural changes increase the returns to advanced degrees, directly explaining the surge in four-year college enrollment and bachelor’s and master’s degree attainment. Additionally, while Amazon’s entry does not induce significant population growth among college-aged individuals (18–24), I find a statistically significant 1.6% increase in the population of individuals holding master’s degrees, suggesting that selective in-migration of already-educated workers contributes partially to the observed gains in advanced degree attainment stocks.

This paper contributes to several strands of distinct literatures. First, it extends research on how local labor market shocks shape educational investment decisions. A robust body of evidence demonstrates that positive shocks increasing returns to low-skill work typically depress educational attainment by raising the opportunity cost of schooling. This pattern has been documented across diverse contexts: housing market booms (Charles et al., 2018), manufacturing expansions (Atkin, 2016), coal mining revivals (Black et al., 2005), and shale oil development (Cascio and Narayan, 2022; Kearney and Wilson, 2018; James, 2017). My findings stand in marked contrast. The educational upgrading I observe suggests that the nature of the shock matters profoundly—specifically, whether the shock is purely low-skill-biased or whether it simultaneously expands opportunities at the upper end of the skill distribution. This suggests that logistics-driven growth can generate human capital dynamics fundamentally different from traditional resource-based booms.

Second, this research informs the literature on place-based economic development policies. While most work in this area evaluates explicit government programs (Busso et al., 2013), Amazon’s expansion represents a qualitatively different phenomenon: a privately-driven location choice heavily shaped by local policy incentives. My findings suggest that attracting such firms can generate positive human capital spillovers, potentially through mechanisms related to productivity externalities and agglomeration economies (Greenstone et al., 2010; Rossi-Hansberg et al., 2021). This provides a novel perspective on the potential long-run benefits of aggressive local recruitment strategies—though a complete welfare analysis would require weighing these gains against the fiscal costs of the incentive packages themselves.

The remainder of the paper proceeds as follows. Section 2.1 provides institutional background on Amazon’s distribution network. Section 2.2 develops the conceptual framework. Section 3.1

describes the data sources and sample construction. Section 3.2 outlines the empirical methodology. Sections 4.1 and 4.2 present the main results on educational attainment and enrollment. Section 4.3 investigates the mechanisms underlying these effects. Section 5 reports robustness checks, and Section 6 concludes and discusses implications for policy.

## 2 Background and Conceptual Framework

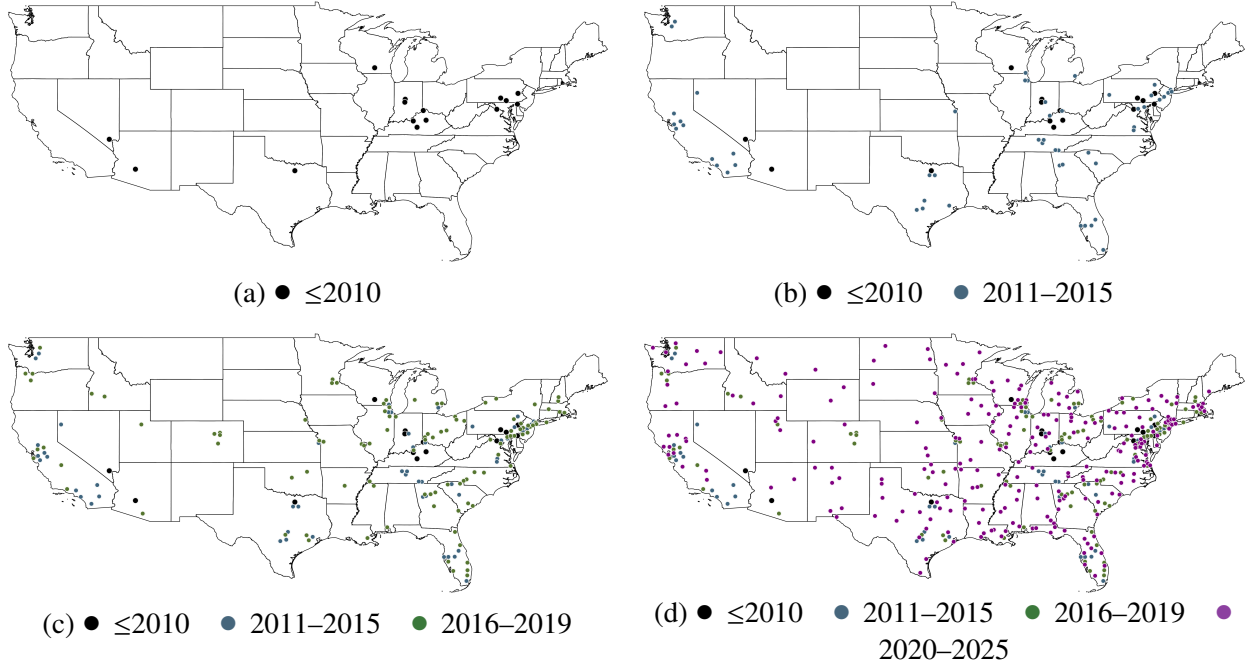
### 2.1 Amazon’s Distribution Network

The rise of e-commerce represents one of the most consequential structural transformations in the U.S. economy over the past two decades. While commonly understood as a digital phenomenon, this transformation rests fundamentally on physical infrastructure: an expansive network of warehouses, sortation facilities, and delivery vehicles that enable rapid fulfillment of online orders. Amazon’s logistics network stands as the paradigmatic example of this infrastructure build-out, serving as the operational foundation for the company’s market dominance in online retail. The period under study—2010 through 2023—encompasses the most dramatic phase of this expansion. As shown in Figure 1, Amazon’s footprint grew from a modest collection of fulfillment centers (FCs) operating in 2010 to a sprawling network exceeding one thousand facilities by the early 2020s.

This geographic expansion followed a deliberate evolution in corporate strategy rather than random market entry. Throughout the 2000s, Amazon’s location decisions were shaped predominantly by sales tax considerations. The company strategically placed facilities in low-population states to avoid triggering sales tax nexus requirements in major consumer markets (Houde et al., 2016), effectively using geographic distance as a tax arbitrage mechanism. However, mounting legal challenges and legislative pressure from state governments rendered this approach increasingly unsustainable. Beginning around 2012, Amazon fundamentally reoriented its location strategy toward operational efficiency and customer proximity (Chung, 2023; MWPVL International, 2025). The revised approach prioritized minimizing shipping costs and delivery times, necessitating construction of a geographically dense network that would inevitably include high-tax, high-population states. This strategic pivot—implemented through staggered market entry over more than a decade—generates the quasi-experimental variation that my empirical analysis exploits.

The contemporary Amazon logistics network operates as a hierarchical, multi-stage system. The fulfillment process originates at large “upstream” fulfillment centers, where inventory is stored and individual orders are picked and packed. Products then flow to “mid-mile” sortation centers for regional consolidation before reaching “last-mile” delivery stations that handle final transportation to customer addresses (Cunningham, 2024). An FC opening—the treatment event examined in this study—thus represents the establishment of a substantial node within this complex, technology-

Figure 1: Staggered Rollout of Amazon’s Fulfillment Center Network by Year, 2005–2025



*Notes:* This figure displays the geographic expansion of Amazon’s fulfillment center network across U.S. counties from 2005 to 2025. Each panel shows the cumulative distribution of facility locations by opening year cohort, illustrating Amazon’s progressive national expansion. Counties are classified by the year their first fulfillment center became operational: black dots indicate facilities opened by 2010, blue dots show facilities opened between 2011 and 2015, green dots represent facilities opened between 2016 and 2019, and purple dots indicate facilities opened between 2020 and 2025. The sequential presentation demonstrates the staggered rollout of Amazon’s distribution network, with initial concentration in major metropolitan areas followed by expansion into secondary markets and broader geographic coverage. Data are compiled from Amazon’s public announcements, press releases, and verified third-party sources tracking the company’s distribution network. The maps use county-level geographic units to show facility locations, though the empirical analysis in this paper is conducted at the commuting zone level to capture integrated local labor markets.

intensive logistics infrastructure.<sup>1</sup>

## 2.2 Conceptual Framework

To provide theoretical structure for the empirical analysis, I develop a stylized model of educational investment under a local labor demand shock. The framework draws on the canonical human capital investment model (Becker, 1964) and adapts the college enrollment framework of Charles et al. (2018) to accommodate two features of the Amazon setting that distinguish it from the resource and manufacturing booms studied in prior work: the income effects generated by rising local wages, and the technology spillovers that accompany a large logistics hub.

Individuals face two sequential educational decisions: whether to complete secondary school and whether to pursue postsecondary education. I index individuals by an academic ability parameter  $\theta_i$  distributed continuously over  $[0, 1]$ .

**The High School Completion Decision.** An individual weighs expected lifetime earnings as a high school graduate,  $Y_{HS}$ , against earnings as a dropout,  $Y_{DO}$ , net of the effort and psychic costs of completing school,  $\kappa$ . The net return to graduation is:

$$R_{HS} = \mathbb{E}[Y_{HS} - Y_{DO}] - \kappa. \quad (1)$$

In standard models of low-skill labor market booms, a positive demand shock raises wages broadly across the lower tail of the skill distribution, compressing the graduation premium  $Y_{HS} - Y_{DO}$  and increasing dropout rates as teenagers substitute toward immediate employment (Black et al., 2005; Charles et al., 2018). Whether the Amazon shock operates through the same channel depends on how the labor demand increase is distributed across the skill distribution. If warehousing wages rise for all low-skill workers regardless of credential, the substitution incentive dominates and high school completion falls. If, however, the wage gains accrue primarily to employed household members rather than to teenagers directly—through the income channel discussed below—the financial pressure to leave school early is relaxed rather than intensified. The net effect on high school completion is therefore theoretically ambiguous and depends on which force dominates.

**The Postsecondary Enrollment Decision.** For high school graduates, I model the college enrollment choice as a selection between educational paths  $c \in \{A, B\}$ , where  $c = A$  denotes two-year

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<sup>1</sup>FC location decisions reflect strategic calculations incorporating local economic conditions. Chung (2023) documents that FCs exhibit higher probability of placement in counties experiencing population density growth but maintaining relatively low median household incomes, consistent with a strategy targeting areas offering large available labor pools proximate to major consumer markets.

colleges and  $c = B$  denotes four-year institutions. Following [Charles et al. \(2018\)](#), the net payoff from enrolling in college type  $c$  for individual  $i$  is:

$$R^c(\theta_i) = \mathbb{E}[\Pi_c] - F_c - \kappa_c(1 - \theta_i) - Y_{NC} \quad (2)$$

where  $\mathbb{E}[\Pi_c]$  is the expected lifetime earnings premium from degree  $c$  relative to immediate labor force entry;  $F_c$  captures direct costs including tuition and fees;  $\kappa_c(1 - \theta_i)$  represents effort costs that decline with ability; and  $Y_{NC}$  is the opportunity cost of college attendance, measured by current wages available to non-college workers. Differentiating with respect to the demand shock yields the central decomposition:

$$\frac{dR^c(\theta_i)}{dS} = \underbrace{\frac{d\mathbb{E}[\Pi_c]}{dS}}_{\text{Upgrading Effect}} - \underbrace{\frac{dY_{NC}}{dS}}_{\text{Substitution Effect}} \quad (3)$$

**The Substitution Effect.** The first force is the standard channel emphasized in prior boom studies. FC entry raises local demand for logistics and warehousing labor, increasing non-college wages ( $dY_{NC}/dS > 0$ ) and thus the opportunity cost of remaining in school. Left unchecked, this mechanism reduces enrollment, consistent with the patterns documented in coal, shale, and housing boom settings ([Black et al., 2005](#); [Cascio and Narayan, 2022](#); [Charles et al., 2018](#)).

**The Upgrading Effect.** The second force operates through the earnings premium term  $\mathbb{E}[\Pi_c]$ . Unlike a resource extraction boom, an FC functions as a high-tech logistics hub that may generate agglomeration spillovers to complementary sectors—attracting suppliers, supporting growth in information-intensive industries, and raising local demand for skilled labor ([Greenstone et al., 2010](#); [Rossi-Hansberg et al., 2021](#)). If FC entry increases the skill premium in the local economy, the expected return to advanced degrees rises ( $d\mathbb{E}[\Pi_B]/dS > 0$ ), creating upward pressure on four-year college enrollment. For two-year colleges, if the skill premium responds minimally to the shock ( $d\mathbb{E}[\Pi_A]/dS \approx 0$ ) while opportunity costs rise, the net effect on community college enrollment is theoretically ambiguous and may be negative.

**The Income Channel.** The model as written treats  $F_c$  as fixed, but there is good reason to think FC entry relaxes this term for marginal students. A substantial literature documents that liquidity constraints are a primary barrier to college enrollment, particularly among low- and middle-income families ([Lochner and Monge-Naranjo, 2011](#); [Lovenheim, 2011](#); [Carneiro and Heckman, 2002](#); [Johnson, 2013](#)). FC entry creates thousands of relatively well-paid jobs, potentially raising household incomes throughout the local economy. To the extent that these income gains accrue to

families with college-aged children, the binding financial constraint on enrollment loosens, generating additional upward pressure on participation beyond what the upgrading effect alone would predict. This income channel operates in parallel with the upgrading effect and reinforces it: both raise the net payoff  $R^c(\theta_i)$ , though through distinct margins of the payoff function.

**A Second Margin: Selective In-Migration.** The model above holds the local population fixed and speaks only to the schooling decisions of individual  $i$  already resident in the commuting zone. But the stock of degree-holders in a commuting zone can rise through a second, distinct margin: already-credentialed workers moving in from elsewhere, drawn by the same demand-side transformation that raises  $\mathbb{E}[\Pi_B]$  above. A worker with a graduate degree who observes a local increase in demand for abstract-task, high-skill labor faces a locational choice analogous to  $i$ 's schooling choice — relocate if the local wage premium net of moving costs exceeds the premium available elsewhere — but this is a decision about *where to live*, not *how much to invest in schooling*, and the two margins carry different observable signatures. This locational margin is well documented in the local labor demand literature: positive demand shocks draw in migrants, with the size and skill-composition of the response an empirical question in its own right (Notowidigdo, 2020; Monras, 2021). Investment via  $R^c(\theta_i)$  requires the time to complete a credential and so can only affect measured attainment with a lag bounded below by program length. Migration requires only the time to relocate, and so can affect the local stock of degree-holders almost immediately upon the shock's onset. This timing distinction is testable and is not a matter of interpretation after the fact: a stock increase that appears within the model's minimum investment lag is more consistent with migration, while one that respects the lag is more consistent with local investment.

The framework generates four predictions that discipline the empirical analysis. First, if the upgrading and income effects together dominate the substitution effect, four-year college enrollment and bachelor's degree attainment should rise following FC entry via local investment, while dropout rates should not increase. Second, the upgrading effect should be concentrated at the four-year margin, where returns to skill are highest, with weaker or null effects on two-year enrollment. Third, the income channel predicts that the enrollment response should be larger in commuting zones where the pre-existing share of credit-constrained households is higher—a prediction I examine through heterogeneity analysis. Fourth, to the extent that any observed increase in advanced-degree attainment operates through the in-migration margin rather than local investment, it should (i) appear faster than the time required to complete the credential, and (ii) be concentrated among individuals who recently changed residence, rather than distributed across the existing population. Whether the net effect on educational attainment is positive or negative is ultimately an empirical question, and one that the staggered rollout of Amazon's network across U.S. commuting zones is well-suited to answer.

## 3 Data and Empirical Strategy

### 3.1 Data Sources and Sample Construction

This study constructs a balanced panel of 170 U.S. commuting zones (CZs) that experienced at least one Amazon facility opening between 2010 and 2023. The analysis integrates multiple administrative and survey-based datasets to examine how Amazon’s local expansion influences educational attainment, college enrollment, migration patterns, and industrial structure. The core datasets include facility-level information from MWPVL International, educational outcomes from the American Community Survey (ACS), enrollment records from the Integrated Postsecondary Education Data System (IPEDS), establishment counts from County Business Patterns (CBP), and labor market indicators from Lightcast.

**Amazon Facility Data.** Information on Amazon’s logistics network is obtained from *MWPVL International, Inc.*, an independent supply chain and logistics consulting firm that maintains comprehensive records of Amazon’s distribution infrastructure. The dataset provides detailed facility-level information including precise warehouse addresses, opening dates, facility types (fulfillment centers, sortation centers, delivery stations, air hubs), and square footage. I exclude Amazon Fresh and Whole Foods locations from the analysis due to their substantially smaller operational scale and more limited regional employment footprint relative to major logistics facilities. Each facility is geocoded and matched to its corresponding commuting zone to construct the treatment timing variable—specifically, the year in which Amazon first established operations within a given local labor market. This facility-entry measure forms the foundation of the staggered treatment design employed in the difference-in-differences analysis.

**Educational Attainment Data.** Educational attainment and demographic data are drawn from the *American Community Survey (ACS)* 1-Year Estimates spanning 2010–2023. Using individual-level microdata obtained through IPUMS, I compute the share of adults aged 25 and older who have completed at least a high school degree, some college coursework, a bachelor’s degree, or a master’s degree. These four attainment categories constitute the primary outcome measures for human capital accumulation. To investigate selective migration as a potential mechanism, I construct migration-adjusted attainment measures by identifying individuals who report having changed residence within the past year, normalized by the total adult population. These variables capture in-migration of educated workers to treated commuting zones. I additionally construct logged population counts disaggregated by age group (18–24 and 25+) and educational category to examine potential demographic composition shifts. All individual-level data are aggregated to the CZ-year

level using ACS person weights and Public Use Microdata Area (PUMA) to CZ geographic cross-walks developed by (Autor and Dorn, 2013).

**College Enrollment Data.** Postsecondary enrollment data are obtained from the *Integrated Postsecondary Education Data System (IPEDS)* for the period 2010–2023. IPEDS provides institution-level annual enrollment counts for all Title IV-eligible postsecondary institutions in the United States, including both public and private colleges and universities. For this analysis, I focus exclusively on public institutions to examine state-subsidized educational investment responses to labor market shocks.

Institution-level enrollment counts are aggregated to the CZ-year level and normalized by the population aged 18–25 to construct enrollment rate measures comparable across markets of varying size. I generate separate enrollment variables for total students as well as disaggregations by gender (male and female), race/ethnicity (White, Black, and Hispanic), and institution type (2-year and 4-year public colleges). These granular measures allow investigation of heterogeneous enrollment responses across demographic groups and educational pathways, providing insight into the flow mechanisms underlying observed changes in attainment stocks.

**Industry Structure Data.** To examine whether Amazon’s entry induces structural transformation in the local economy, I utilize establishment count data from the *County Business Patterns (CBP)* dataset for 2014–2023. CBP, compiled annually by the U.S. Census Bureau, provides comprehensive counts of business establishments classified by detailed North American Industry Classification System (NAICS) codes at the county level. I aggregate county-level establishment counts to the commuting zone level using county-to-CZ geographic concordances.

The analysis focuses on three key establishment measures: total establishments across all sectors, establishments in the information sector (NAICS 51), and establishments in the manufacturing sector (NAICS 31-33). The information sector captures technology-intensive industries including software publishers, data processing services, and telecommunications—sectors that may experience positive spillovers from Amazon’s technology-driven operations. Manufacturing establishments provide a comparison sector less directly linked to e-commerce logistics. These measures allow assessment of whether Amazon entry catalyzes growth in complementary high-skill sectors beyond direct logistics employment.

**Labor Market and Task Content Data.** Finally, I incorporate real-time labor demand data from *Lightcast* (formerly known as Emsi Burning Glass) covering 2014–2023. As documented in [Andreadis et al. \(2025\)](#), Lightcast aggregates online job postings from over 40,000 sources including corporate websites, job boards, and newspapers, providing comprehensive coverage of advertised

labor demand across geographic areas and occupations.

From these data, I construct two sets of measures. First, I calculate the share of job postings related to artificial intelligence and advanced technology occupations, capturing shifts toward high-skill digital employment. Second, I construct measures of local task composition following the methodology in [Autor and Dorn \(2013\)](#), classifying occupations into abstract (cognitive/analytical), routine (procedural/repetitive), and manual (physical) task categories. These occupation-level task intensities are weighted by local employment shares within each CZ-year cell to generate aggregate task composition measures. These variables enable investigation of mechanisms linking Amazon’s entry to changes in labor demand structure, specifically testing whether FCs generate spillovers that increase demand for high-skill, abstract-task labor as hypothesized in the conceptual framework.

Collectively, these data sources provide a comprehensive empirical foundation for analyzing how large-scale logistics investments affect local human capital formation and economic structure. The integration of survey-based outcomes (ACS), administrative enrollment records (IPEDS), establishment dynamics (CBP), and real-time labor demand indicators (Lightcast) enables examination of both short-run flow adjustments and longer-run stock changes in educational attainment.

## 3.2 Empirical Strategy

I employ a staggered difference-in-differences design to estimate the causal effect of Amazon fulfillment center entry on local educational outcomes. The analysis is conducted at the commuting zone level, which represents natural local labor markets and is essential for capturing regional spillover effects that may extend beyond individual county boundaries.

Following [Cunningham \(2024\)](#) and [Chung \(2023\)](#), treatment is defined as Amazon’s entry into a commuting zone, measured by the year of the first fulfillment center or warehouse opening. To account for potential anticipation effects, I define a commuting zone as being treated one year prior to the first operational date of its fulfillment center.<sup>2</sup> My treatment indicator therefore captures both anticipatory responses and the realized effects of FC presence. The effects I measure reflect the cumulative impact of all Amazon facilities operating in a commuting zone, rather than isolating the impact of just the initial establishment.

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<sup>2</sup>Anticipation effects can occur as local residents and markets react to the announcement of a new FC before it becomes operational. This can include pre-opening hiring for construction and management. Supporting this timeframe, [Chung \(2023\)](#) finds that the average period between an FC’s announcement and the actual start of its operations is approximately six months. A one-year window therefore provides a reasonable period to capture these anticipatory dynamics.

### 3.2.1 Baseline Specification

The baseline estimating equation follows the standard two-way fixed effects form:

$$Y_{ct} = \theta_t + \alpha_c + \beta \cdot FC_{ct} + \varepsilon_{ct} \quad (4)$$

where  $Y_{ct}$  denotes the educational outcome in commuting zone  $c$  in year  $t$ ;  $\alpha_c$  represents commuting zone fixed effects that absorb time-invariant local characteristics;  $\theta_t$  are year fixed effects controlling for aggregate time trends common across all regions; and  $FC_{ct}$  is an indicator equal to one if Amazon operates a fulfillment center in commuting zone  $c$  at year  $t$ , and zero otherwise. The coefficient  $\beta$  captures the average treatment effect of Amazon entry on the outcome by averaging the treatment effects for each cohort across all post-treatment years, and then averaging these effects across all cohorts.

### 3.2.2 Event-Study Specification

To examine how treatment effects evolve over time and to assess the plausibility of the parallel trends assumption, I estimate an event-study specification:

$$Y_{ct} = \theta_t + \alpha_c + \sum_{j=-k}^n \beta_j \cdot D_{c,t-j} + \varepsilon_{ct} \quad (5)$$

where  $D_{c,t-j}$  is an indicator variable equal to one if Amazon entry occurred  $j$  years ago in commuting zone  $c$ , and zero otherwise. I omit  $j = -1$  as the reference period. The coefficients  $\{\beta_j\}$  trace out the dynamic treatment effect at each event time  $j$  and capture event-time average treatment effects on the treated (ATTs). Pre-treatment coefficients ( $j < -1$ ) provide a visual test of parallel trends, while post-treatment coefficients ( $j \geq 0$ ) reveal how treatment effects evolve over time.

### 3.2.3 Identification Strategy

Identification relies on variation in the timing of Amazon's entry across commuting zones. The identifying assumption is parallel trends: absent Amazon entry, treated and not-yet-treated commuting zones would have followed the same trajectory in educational outcomes. This assumption is not directly testable, but two pieces of evidence bear on it. The event-study specification in equation (5) shows flat, statistically insignificant pre-treatment coefficients (Section 4.1 below). More formally, I test the moment restrictions parallel trends implies with balance tests on pre-treatment first-differences following [Marcus and Sant'Anna \(2021\)](#); Appendix Table A4 (also Online Appendix Table A5) shows that for the vast majority of outcomes and treatment cohorts, I cannot reject the null of balanced pre-treatment trends between treated and not-yet-treated commuting zones.

The choice of control group matters for this test to mean anything. Amazon’s site-selection decisions are strategic and non-random ([Chung, 2023](#)): commuting zones that never receive a facility differ systematically, in levels, from those that eventually do. Comparing treated zones only to other eventually-treated zones, the not-yet-treated design used throughout this paper, sidesteps that selection-on-levels problem by exploiting timing variation among units similar enough to receive a facility eventually.

A harder problem is whether FC placement is correlated not just with pre-treatment levels but with unobserved local growth trajectories: an early Amazon entrant could simply be a commuting zone already on a rising path, for reasons that also happen to make it attractive to Amazon. [Atkin \(2016\)](#) confronts the same problem in a comparable setting, export-manufacturing plant openings across Mexican municipalities at roughly the commuting-zone scale, and addresses it with an instrument for firm location built from cross-market variation in firms’ national expansion plans. No comparable instrument is available for Amazon’s FC siting, since there is no external, plausibly exogenous source of variation in a commuting zone’s likelihood of receiving a facility that is unrelated to its own economic trajectory. Absent an instrument, this paper’s defense against selection on trends is the balance tests and flat pre-trends above, which test that restriction directly rather than assume it. They do not rule out every time-varying confound: a rezoning decision or a regional logistics-cluster expansion coincident with FC entry could also move educational outcomes independently of Amazon (a limitation discussed further in [Section 6](#)). They are, however, the best direct evidence available on the specific threat an instrument would otherwise address.

The mechanism results in [Section 4.3](#) offer a second, independent check, this time on the content of the shock rather than its timing: growth concentrated in the information sector rather than other industries, employment shifting toward abstract-cognitive tasks rather than routine or manual ones, and technology job postings rising in step. A generic confound, such as a rezoning decision or an unrelated regional boom, has no obvious reason to produce that same combination of changes at that same point in each commuting zone’s event time. Amazon’s own mechanism, attracting complementary high-skill firms, does. That is why the two checks reinforce each other rather than simply repeating the same test twice.

### **3.2.4 Addressing Heterogeneous Treatment Effects**

Recent econometric research demonstrates that standard two-way fixed effects estimators can produce misleading estimates in staggered adoption settings when treatment effects are heterogeneous across units or over time ([Goodman-Bacon, 2021](#); [Sun and Abraham, 2021](#)). The issue arises because TWFE estimators implicitly use already-treated units as controls for newly-treated units, potentially generating “negative weights” that bias estimates.

To address these concerns, I implement the [Callaway and Sant’Anna \(2021\)](#) estimator, which is

robust to heterogeneous treatment effects in staggered designs. This approach constructs treatment effect estimates by comparing each newly-treated cohort to not-yet-treated units only, avoiding contamination from already-treated observations. The procedure aggregates cohort-specific average treatment effects on the treated (ATTs) into overall treatment effects and event-study parameters while explicitly accounting for treatment timing variation.

This estimator provides three key advantages. First, it produces unbiased estimates even when treatment effects vary by cohort or evolve dynamically over time. Second, it allows flexible aggregation of group-time specific effects into interpretable summary parameters. Third, it enables valid inference through a wild bootstrap procedure that accounts for clustering and estimation uncertainty.

### 3.2.5 Control Variables

I do not include time-varying control variables in my baseline specifications. Following [Callaway and Sant’Anna \(2021\)](#), time-varying controls can only be included if they are independent of treatment. However, I view nearly all local economic and demographic variables as potentially affected by Amazon’s entry. Including variables that are themselves affected by treatment would lead to post-treatment bias, where the estimated treatment effect would capture only the direct effect of Amazon entry not mediated through the control variables, rather than the total causal effect of interest.

Additionally, I show that treated and not-yet-treated commuting zones are balanced with respect to pre-treatment levels and changes in key demographic characteristics, including age, race, ethnicity, foreign-born status, and family structure. This balance provides further support for the parallel trends assumption and reduces concerns about omitted variable bias. During estimation, I use the “doubly robust” version of the [Callaway and Sant’Anna \(2021\)](#) estimator, and standard errors are clustered at the Commuting Zone (CZ) level and calculated using the recommended bootstrap procedure.

## 4 Results

This section presents evidence on how Amazon fulfillment center entry affects educational outcomes in local labor markets. I begin by documenting effects on educational attainment stocks among adults before examining enrollment flows into higher education. The results reveal a consistent pattern of educational upgrading across both margins, contrasting sharply with the substitution effects documented in prior resource boom literature. Table 1 reports summary statistics for the main outcome variables used throughout the analysis.

Table 1: Summary Statistics

| Variable                                                                | Mean      | SD       | N     |
|-------------------------------------------------------------------------|-----------|----------|-------|
| <i>Panel A. Educational Attainment (% of adults 25+)</i>                |           |          |       |
| High School+                                                            | 88.43     | 4.27     | 2,097 |
| Some College+                                                           | 59.87     | 6.85     | 2,097 |
| Bachelor's+                                                             | 28.88     | 6.48     | 2,097 |
| Master's+                                                               | 10.84     | 3.35     | 2,097 |
| <i>Panel B. 4-Year Public College Enrollment Rate (% of ages 18–25)</i> |           |          |       |
| Total Enrollment                                                        | 15.15     | 18.25    | 2,268 |
| Male Enrollment                                                         | 6.76      | 8.30     | 2,268 |
| Female Enrollment                                                       | 8.39      | 10.14    | 2,268 |
| <i>Panel C. Income and Poverty</i>                                      |           |          |       |
| Median Household Income (\$)                                            | 49,051.89 | 9,104.28 | 1,410 |
| Poverty Rate                                                            | 0.144     | 0.037    | 1,410 |

*Notes:* This table reports pre-treatment means, standard deviations, and sample sizes for the main outcome variables analyzed in this paper, at the commuting zone-year level. Panel A: share of adults aged 25 and older who have completed at least the specified level of education (ACS 1-Year estimates, 2010–2023). Panel B: share of individuals aged 18–25 enrolled in four-year public colleges (IPEDS, 2010–2023, denominators from ACS). Panel C: median household income and poverty rate (ACS 1-Year estimates, 2014–2023). The estimation sample consists of a balanced panel of 170 commuting zones that eventually receive Amazon facilities during the study period.

## 4.1 Educational Attainment

I first provide descriptive evidence on educational composition changes surrounding Amazon entry. Online Appendix Figure A1 compares average educational distributions in the five years before and after an FC opens within a commuting zone.<sup>3</sup> Panel A of that figure establishes the pre-treatment baseline; Panel B displays percentage changes between the five-year pre- and post-entry windows, offering initial suggestive evidence of upgrading: the share lacking high school credentials declines 2.0%, while shares with some college (+2.9%), bachelor's degrees (+7.9%), and master's degrees (+8.8%) all increase.

<sup>3</sup>The pre-treatment baseline reported here (87.6% with at least a high school diploma, 32.4% with a bachelor's degree or higher) differs slightly from Table 1's and Table 2's pre-treatment means (88.43%, 28.88%) because the two are constructed differently: this figure restricts to a fixed five-year window ( $t = -5$  to  $t = -1$ ) around each commuting zone's own treatment year, while the regression tables' pre-treatment mean pools across all available pre-treatment commuting zone-year observations in the estimation sample – which varies in length across cohorts under staggered adoption, since early-treated cohorts (e.g., 2014) contribute fewer pre-periods than late-treated cohorts (e.g., 2023). Both are legitimate descriptions of the same underlying data; they answer slightly different questions (a fixed-window comparison versus a pooled panel mean).

**Validating Parallel Trends.** Before presenting causal estimates, I assess the identifying assumption’s plausibility. The staggered difference-in-differences framework requires parallel trends in first-differences between treated and not-yet-treated groups prior to treatment.<sup>4</sup> Appendix Tables A4 and A5 present formal balance tests for educational outcomes and demographic covariates across treatment cohorts.

Results provide mixed but ultimately supportive evidence. For cohorts treated from 2017 onward (Cohorts 2 and 3), pre-treatment trends are well-balanced across nearly all educational variables. However, the earliest cohort (2014–2016) exhibits some pre-existing differential trends, particularly for bachelor’s and master’s attainment. Importantly, key at-risk populations (ages 18–24) appear balanced across all cohorts, and the event-study plots presented below show flat, statistically insignificant pre-trends in years immediately preceding treatment for the full sample. This visual evidence, combined with balance in later cohorts, supports causal interpretation of the findings.

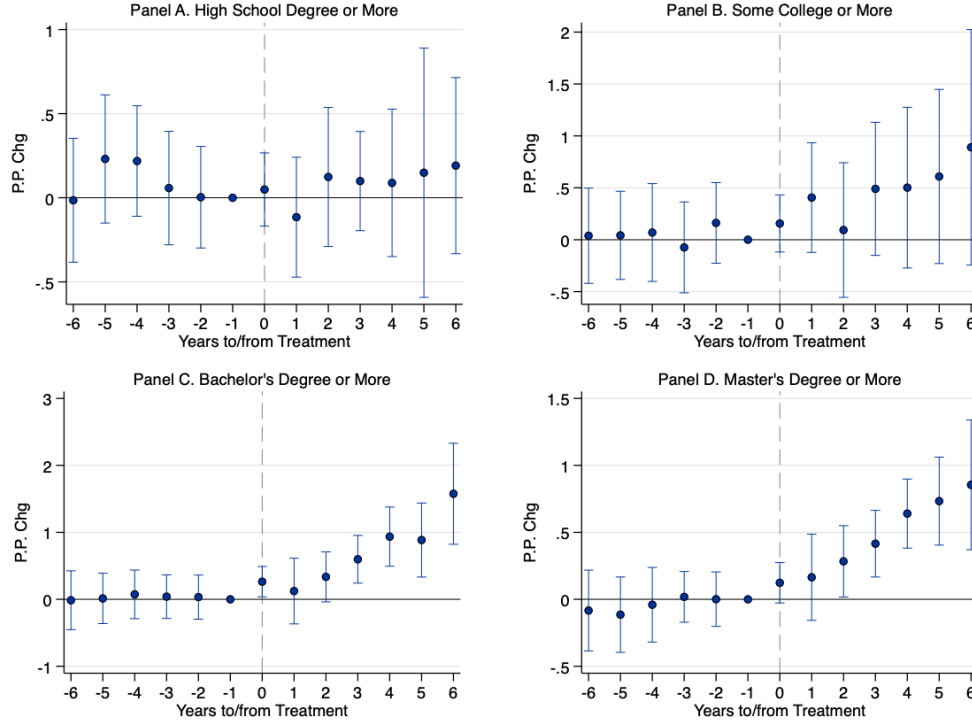
**Main Results: Event-Study Evidence.** Figure 2 presents dynamic treatment effects on educational attainment. Each panel plots event-time coefficients from the Callaway-Sant’Anna estimator, with year  $-1$  normalized to zero. Pre-treatment coefficients (years  $-6$  to  $-2$ ) cluster tightly around zero across all specifications, providing visual confirmation of parallel pre-trends. Following Amazon entry at year zero, attainment trajectories diverge sharply upward for higher education categories.

Panels A and B show positive but modest, statistically insignificant effects on high school completion and some college attainment. The pattern changes for four-year degrees. Panel C shows bachelor’s degree attainment rising from treatment onward, reaching statistical significance around three years post-entry and continuing to grow thereafter. The early movement is faster than a pure enrollment-to-completion pipeline alone would generate — enrollment itself, in Section 4.2, does not move until roughly year two — so some of it may reflect other margins, such as returning students or workers close to finishing a degree already in progress; the significance at year three is nonetheless consistent with a meaningful share of the response coming from new completions. The master’s degree response (Panel D) moves faster still, reaching statistical significance within two years and growing steadily through the end of the observation window. That timing is hard to reconcile with a graduate degree completed from a standing start, which is why I turn next to population change as an alternative explanation.

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<sup>4</sup>As discussed in [Marcus and Sant’Anna \(2021\)](#), balance in first-differences constitutes a direct test of moment restrictions implied by parallel trends.

Figure 2: Event-Study Estimates: Effects of Amazon’s Entry on Educational Attainment



*Notes:* This figure presents dynamic estimates of Amazon fulfillment center entry on educational attainment levels within commuting zones. The dependent variables measure percentage-point changes in the share of adults aged 25 and older who have completed at least the specified level of education: high school diploma or equivalent, some college education, bachelor’s degree, or master’s degree. Data are from the American Community Survey (ACS) 1-Year estimates for 2010–2023, constructed from ACS microdata and aggregated to the commuting zone level using person weights. The estimation sample consists of a balanced panel of commuting zones that eventually receive Amazon facilities during the study period. The vertical dashed line indicates the year of Amazon’s first fulfillment center opening in each commuting zone, with year  $t = -1$  serving as the omitted reference period. Estimates are obtained from a staggered difference-in-differences specification following [Callaway and Sant’Anna \(2021\)](#), which compares treated commuting zones to not-yet-treated control units while incorporating commuting zone fixed effects and year fixed effects. Each point represents an event-time average treatment effect on the treated (ATT), with vertical bars indicating pointwise 95% confidence intervals (point estimate  $\pm 1.96$  times the wild bootstrap standard error), clustered at the commuting zone level.

Table 2 quantifies average treatment magnitudes by aggregating group-time effects into overall ATT parameters. Consistent with event-study patterns, effects concentrate in higher education. Bachelor’s degree attainment increases 0.58 percentage points (significant at 1%), while master’s attainment rises 0.41 percentage points (significant at 1%). High school completion shows a modest, marginally significant increase of 0.20 percentage points, while the some-college effect remains positive but imprecise. These magnitudes represent economically meaningful shifts: the bachelor’s effect equals approximately 2% of the pre-treatment mean, while the master’s effect represents

nearly 4% of baseline attainment.

Table 2: Effects of Amazon’s Entry on Educational Attainment

|                | High School+      | Some College+    | Bachelor’s+         | Master’s+           |
|----------------|-------------------|------------------|---------------------|---------------------|
| Amazon Entry   | 0.202*<br>(0.113) | 0.374<br>(0.278) | 0.575***<br>(0.154) | 0.408***<br>(0.113) |
| Mean Dep. Var. | 88.43             | 59.87            | 28.88               | 10.84               |
| SD             | 4.27              | 6.85             | 6.48                | 3.35                |
| N              | 2,097             | 2,097            | 2,097               | 2,097               |

*Notes:* This table presents average treatment effects on the treated (ATT) for the impact of Amazon fulfillment center entry on educational attainment within commuting zones. The dependent variables measure the percentage-point share of adults aged 25 and older who have completed at least the specified level of education: high school diploma or equivalent, some college education, bachelor’s degree, or master’s degree. Data are from the American Community Survey (ACS) 1-Year estimates for 2010–2023, aggregated to the commuting zone level using person weights. The estimation sample consists of a balanced panel of 2,097 commuting zone-year observations from commuting zones that eventually receive Amazon facilities during the study period. Estimates are obtained from a staggered difference-in-differences specification following [Callaway and Sant’Anna \(2021\)](#), which compares treated commuting zones to not-yet-treated control units while incorporating commuting zone fixed effects and year fixed effects. Standard errors, reported in parentheses, are clustered at the commuting zone level using wild bootstrap procedures. Statistical significance levels: \*  $p < 0.10$ , \*\*  $p < 0.05$ , \*\*\*  $p < 0.01$ . Mean Dep. Var. reports the pre-treatment mean of the dependent variable in treated commuting zones. SD denotes the standard deviation of the dependent variable.

The master’s-degree result in particular raises an immediate question, motivated by the conceptual framework’s timing prediction above: are these gains coming from residents investing more in their own education, or is Amazon simply drawing already-credentialed people in from elsewhere? I separate the two using a two-way fixed effects model of log population by education group,

$$\ln(\text{Population}_{ct}) = \alpha_c + \gamma_t + \beta \text{Post}_{ct} + \varepsilon_{ct}, \quad (6)$$

where  $\alpha_c$  and  $\gamma_t$  are commuting zone and year fixed effects and standard errors are clustered at the CZ level.

Table 3 shows no detectable change in total population or in the 18–24 age group — no broad demographic inflow, and no sign that the college-age population itself is being replaced by newcomers. The one exception is master’s degree holders, whose population rises by 1.6% (significant at 5%). Read alongside the master’s-degree attainment result reaching significance within roughly two years of treatment (Panel D above) — too fast to reflect new local completion of a graduate degree, per the timing logic in Section 2.2 — the more natural reading is that the master’s-degree

attainment result is substantially, if not entirely, a selective in-migration effect rather than local investment. The stability of the 18–24 group and the bachelor’s-degree timing (Panel C, reaching significance only after roughly three years, consistent with new enrollment and completion) point the other way for the bachelor’s and enrollment results, which look like genuine local investment.

Table 3: Results: Local Population and Selective Migration

|                | (1) 18–24        | (2) High School  | (3) Some College | (4) Bachelor’s   | (5) Master’s       |
|----------------|------------------|------------------|------------------|------------------|--------------------|
| Amazon Entry   | 0.001<br>(0.006) | 0.008<br>(0.006) | 0.008<br>(0.006) | 0.008<br>(0.006) | 0.016**<br>(0.007) |
| Mean Dep. Var. | 10.99            | 11.52            | 11.60            | 11.05            | 10.17              |
| N              | 2,096            | 2,096            | 2,096            | 2,096            | 2,096              |

*Notes:* This table presents two-way fixed effects estimates of Amazon fulfillment center entry on local population levels by education group. The dependent variable in each column is the natural logarithm of population for the specified demographic group within a commuting zone. Each column reports results from a separate regression of log population on an Amazon entry indicator, incorporating commuting zone fixed effects and year fixed effects. Data are from the American Community Survey (ACS) 1-Year estimates covering the period 2010–2023. The estimation sample includes commuting zone-year observations for areas that eventually receive Amazon facilities. Standard errors, reported in parentheses, are clustered at the commuting zone level. Statistical significance levels: \*  $p < 0.10$ , \*\*  $p < 0.05$ , \*\*\*  $p < 0.01$ . Mean Dep. Var. reports the pre-treatment mean of log population in treated commuting zones.

**Heterogeneity by Gender.** I next investigate whether upgrading effects vary across demographic groups. Online Appendix Figure A2 and Table A1 display event-study estimates and aggregated ATTs disaggregated by gender for bachelor’s and master’s attainment. Bachelor’s degree effects reach 0.58 percentage points for men (significant at 1%) and 0.55 percentage points for women (significant at 1%), while master’s effects equal 0.45 percentage points for men (significant at 1%) and 0.36 percentage points for women (significant at 1%). The similarity in point estimates and statistical significance across genders indicates educational upgrading represents a broad-based phenomenon rather than a group-specific response.

**High School Dropout Rates.** A crucial prediction distinguishing the upgrading hypothesis from traditional substitution effects concerns high school dropout behavior. Prior literature on resource booms documents increased dropout rates as improved low-skill labor market opportunities raise the opportunity cost of schooling (Black et al., 2005; Charles et al., 2018). Appendix Figure A1 examines whether similar patterns emerge following Amazon entry by plotting event-study estimates for the share of 16–19 year-olds lacking high school diplomas.

Pre-treatment coefficients exhibit no systematic trend, supporting parallel trends. Post-treatment estimates cluster near zero throughout the observation window, with none approaching statistical

significance. The overall ATT equals  $-0.15$  percentage points with a standard error of  $0.31$ , statistically indistinguishable from zero. This null finding contrasts sharply with prior boom literature and provides critical evidence that Amazon’s labor demand shock differs fundamentally from traditional low-skill employment expansions.

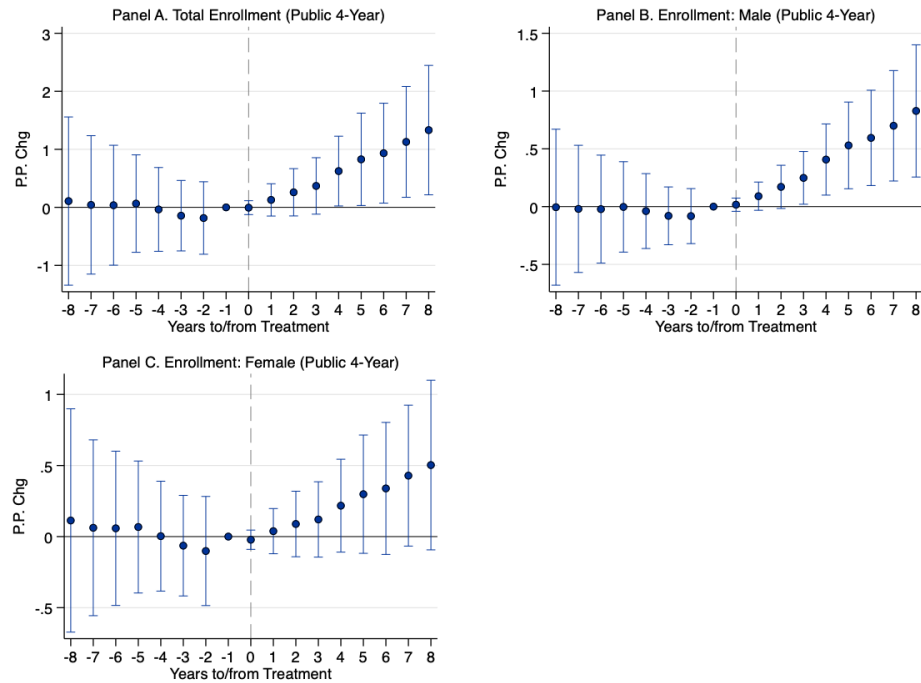
## 4.2 College Enrollment

Having documented increases in educational attainment stocks, I now examine enrollment flows into postsecondary institutions. These flow measures illuminate the mechanisms through which attainment shifts occur and test whether observed effects reflect genuine human capital investment or alternative channels such as selective migration.

**Four-Year Public College Enrollment.** Figure 3 displays event-study estimates for enrollment rates in four-year public institutions. Pre-treatment coefficients (years  $-8$  to  $-2$ ) remain flat and statistically insignificant across all panels, again supporting parallel trends. Following treatment, Panel A reveals a clear positive trajectory in total enrollment, becoming statistically significant approximately four years post-entry. Panels B and C disaggregate by gender, showing similar positive post-treatment patterns for both male and female enrollment, though male effects appear somewhat larger in magnitude.

Table 4 quantifies these effects through aggregated ATT estimates. Total four-year enrollment increases  $0.52$  percentage points (significant at  $5\%$ ), representing approximately  $3.4\%$  of the pre-treatment mean. Disaggregation by gender reveals that male enrollment drives this overall effect, rising  $0.33$  percentage points (significant at  $1\%$ ), while the female effect of  $0.18$  percentage points remains positive but imprecise. These enrollment increases provide a direct mechanism linking Amazon entry to the observed bachelor’s degree attainment gains documented above.

Figure 3: Event-Study Estimates: Effects of Amazon’s Entry on Enrollment in 4-Year Public Colleges



*Notes:* This figure presents dynamic estimates of Amazon fulfillment center entry on enrollment rates in four-year public colleges within commuting zones. The dependent variables measure the percentage-point share of individuals aged 18–25 enrolled in four-year public institutions, constructed by dividing total enrollment in four-year public colleges by the population aged 18–25 in each commuting zone and year. Enrollment data are from the Integrated Postsecondary Education Data System (IPEDS) for 2010–2023, aggregated from institution-level observations to the commuting zone-year level. Population data for the denominator are from the American Community Survey (ACS) 1-Year estimates. The estimation sample consists of a balanced panel of commuting zones that eventually receive Amazon facilities during the study period. The vertical dashed line indicates the year of Amazon’s first fulfillment center opening in each commuting zone, with year  $t = -1$  serving as the omitted reference period. Estimates are obtained from a staggered difference-in-differences specification following [Callaway and Sant’Anna \(2021\)](#), which compares treated commuting zones to not-yet-treated control units while incorporating commuting zone fixed effects and year fixed effects. Each point represents an event-time average treatment effect on the treated (ATT), with vertical bars indicating pointwise 95% confidence intervals (point estimate  $\pm 1.96$  times the wild bootstrap standard error), clustered at the commuting zone level.

Table 4: Effects of Amazon’s Entry on Enrollment in 4-Year Public Colleges

|                | Total Enrollment   | Male Enrollment     | Female Enrollment |
|----------------|--------------------|---------------------|-------------------|
| Amazon Entry   | 0.515**<br>(0.250) | 0.333***<br>(0.122) | 0.182<br>(0.134)  |
| Mean Dep. Var. | 15.15              | 6.76                | 8.39              |
| SD             | 18.25              | 8.30                | 10.14             |
| N              | 2,268              | 2,268               | 2,268             |

*Notes:* This table presents average treatment effects on the treated (ATT) for the impact of Amazon fulfillment center entry on enrollment rates in four-year public colleges within commuting zones. The dependent variables measure the percentage-point share of individuals aged 18–25 enrolled in four-year public institutions, disaggregated by gender. Enrollment data are from the Integrated Postsecondary Education Data System (IPEDS) for 2010–2023, aggregated from institution-level observations to the commuting zone-year level and divided by the population aged 18–25 (from ACS 1-Year estimates) to construct enrollment rates. The estimation sample consists of a balanced panel of 2,268 commuting zone-year observations from commuting zones that eventually receive Amazon facilities during the study period. Estimates are obtained from a staggered difference-in-differences specification following [Callaway and Sant’Anna \(2021\)](#), which compares treated commuting zones to not-yet-treated control units while incorporating commuting zone fixed effects and year fixed effects. Standard errors, reported in parentheses, are clustered at the commuting zone level using wild bootstrap procedures. Statistical significance levels: \*  $p < 0.10$ , \*\*  $p < 0.05$ , \*\*\*  $p < 0.01$ . Mean Dep. Var. reports the pre-treatment mean of the dependent variable in treated commuting zones. SD denotes the standard deviation of the dependent variable.

Racial heterogeneity in four-year enrollment responses appears substantial. Online Appendix Figure A3 and Table A2 disaggregate effects across White, Black, and Hispanic students. The strongest post-treatment increases occur among White students, while effects for Black and Hispanic students remain smaller and less precisely estimated. The aggregated ATT for White students equals 0.37 percentage points (significant at 5%), while estimates for Black and Hispanic students equal 0.06 percentage points for each group (neither statistically significant). These differential responses warrant further investigation in future work but fall outside this paper’s primary scope.

**Two-Year Public College Enrollment.** Traditional substitution effect theories predict that improved low-skill labor market opportunities should reduce enrollment in lower-return educational programs, particularly community colleges. Online Appendix Figures A4 and A5 present event-study estimates for two-year public college enrollment disaggregated by gender and by race. Contrary to substitution predictions, both reveal modest positive (though largely insignificant) post-treatment trends rather than declines.

Online Appendix Table A3 confirms these visual impressions through aggregated ATT esti-

mates. All point estimates are positive but imprecise: male enrollment rises 0.23 percentage points (standard error 0.169), female enrollment increases 0.25 percentage points (standard error 0.185), and racial subgroup estimates range from 0.03 to 0.42 percentage points with correspondingly large standard errors. The consistent absence of negative effects provides further evidence against traditional substitution mechanisms.

Collectively, the enrollment evidence strongly reinforces the attainment findings. Significant increases in four-year college enrollment provide a clear flow mechanism explaining observed gains in bachelor's and master's degree attainment. The absence of negative effects on community college enrollment, combined with modest positive point estimates, aligns with the positive effects on some college attainment. These patterns indicate Amazon entry generates an upgrading shock that stimulates human capital investment rather than a substitution shock that diverts students from educational pathways.

### 4.3 Mechanisms

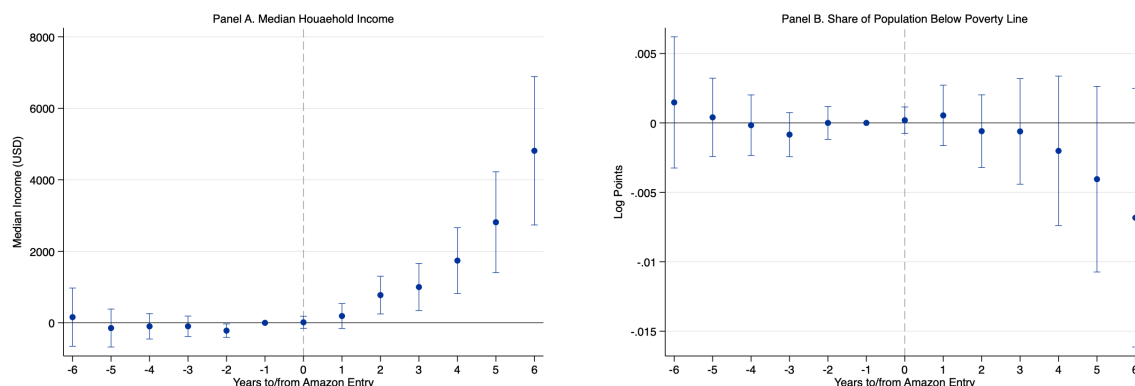
The attainment results run against what the boom literature would predict. Coal booms, housing booms, the fracking boom — in each case cheaper labor market opportunities pull teenagers toward work and away from school (Black et al., 2005; Charles et al., 2018; Cascio and Narayan, 2022). Amazon's entry does not fit this pattern. Dropout rates do not move, high school completion ticks up modestly, and four-year enrollment and degree attainment rise substantially. This section traces that divergence to the two channels laid out in Section 2.2: an income channel that loosens the financial constraint on staying in school, and a demand-side spillover that reshapes the local economy toward skilled work.

**The Income Channel: Income and Poverty.** If rising household income relaxes credit constraints on schooling, the substitution effect emphasized in prior boom studies should be muted here. Figure 4 and Table 5 speak directly to this. Median household income climbs steadily after FC entry, reaching an average gain of \$982 (significant at 1%), while the share of the population below the poverty line falls by 0.25 percentage points over the same window. Neither series moves before treatment.

A large literature on credit constraints and college-going treats income shocks of this kind as directly relevant to enrollment decisions (Lochner and Monge-Naranjo, 2011; Lovenheim, 2011; Carneiro and Heckman, 2002; Johnson, 2013): families on the margin of affording tuition, or of forgoing a teenager's earnings, respond to exactly this kind of income gain. That gives a plausible account of why high school completion edges up without any corresponding rise in dropout — the wage gains from FC entry appear to flow mainly to working-age household members rather than to

teenagers themselves, easing the pressure to leave school rather than intensifying it.

Figure 4: Dynamic Effects of Amazon Entry on Local Income and Poverty Levels



*Notes:* This figure presents dynamic estimates of Amazon fulfillment center entry on median household income and poverty rates within commuting zones. Panel A displays effects on median household income (measured in dollars), while Panel B shows effects on the poverty rate (measured as the percentage-point share of the population below the federal poverty line). Data are from the American Community Survey (ACS) 1-Year estimates for 2014–2023, aggregated to the commuting zone level using person weights. The estimation sample consists of a balanced panel of commuting zones that eventually receive Amazon facilities during the study period. The vertical dashed line indicates the year of Amazon’s first fulfillment center opening in each commuting zone, with year  $t = -1$  serving as the omitted reference period. Estimates are obtained from a staggered difference-in-differences specification following [Callaway and Sant’Anna \(2021\)](#), which compares treated commuting zones to not-yet-treated control units while incorporating commuting zone fixed effects and year fixed effects. Each point represents an event-time average treatment effect on the treated (ATT), with vertical bars indicating pointwise 95% confidence intervals (point estimate  $\pm 1.96$  times the wild bootstrap standard error), clustered at the commuting zone level.

**The Demand-Side Spillover Channel: Changes in Establishments.** An FC is not just a warehouse. If its arrival pulls in suppliers and complementary firms, the local economy should tilt toward sectors that hire skilled workers — exactly the mechanism behind agglomeration spillovers documented for large plant openings ([Greenstone et al., 2010](#)). Appendix Figure A2 and Table A1 test this. Total establishments grow by 1.9% (significant at 5%) following entry, and the information sector alone grows by 2.6% (significant at 10%) — a sector with little direct connection to logistics but a strong claim on educated labor. Manufacturing, by contrast, shows no discernible movement.<sup>5</sup>

<sup>5</sup>[Greenstone et al. \(2010\)](#) find that a large manufacturing plant opening raises incumbent firm productivity by 12% in the same county, with the largest gains among firms sharing labor and technology pools.

Table 5: Effects of Amazon’s Entry on Median Household Income and Poverty Rate

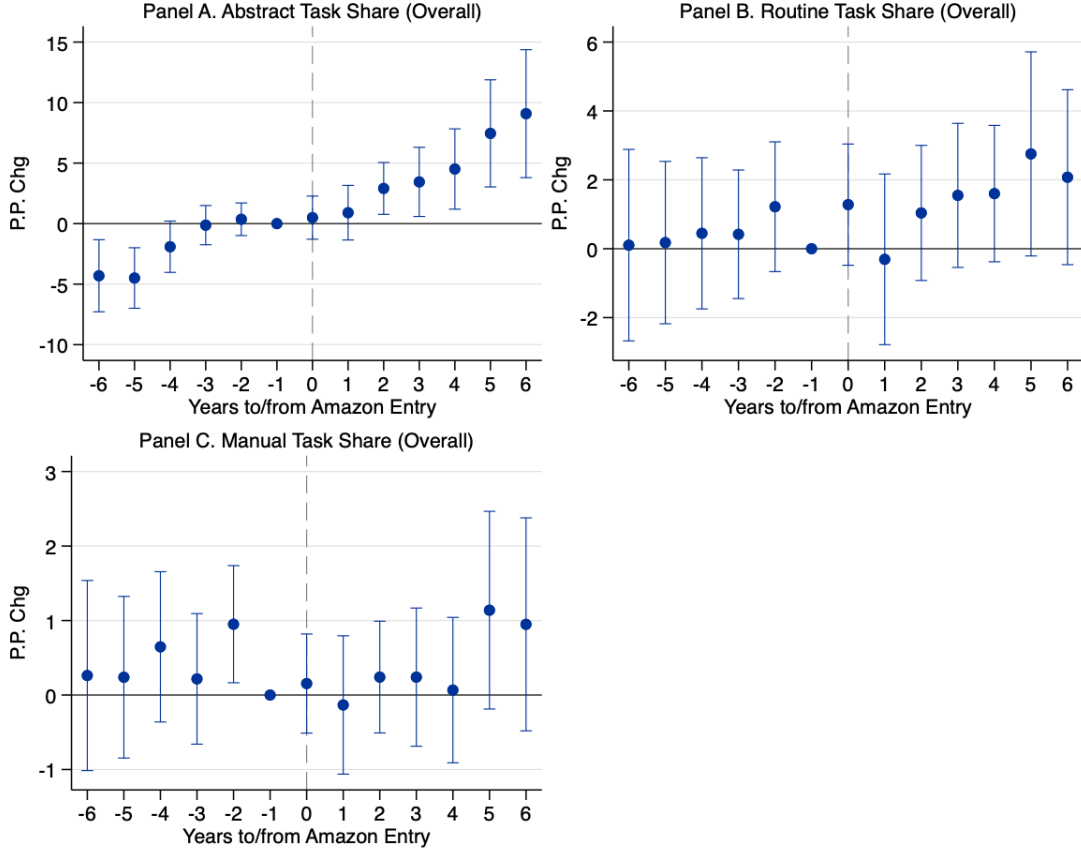
|                | Median Household Income (\$) | Poverty Rate           |
|----------------|------------------------------|------------------------|
| Amazon Entry   | 982.16***<br>(229.95)        | −0.0025***<br>(0.0009) |
| Mean Dep. Var. | 49,051.89                    | 0.144                  |
| SD             | 9,104.28                     | 0.037                  |
| N              | 1,410                        | 1,410                  |

*Notes:* This table presents average treatment effects on the treated (ATT) for the impact of Amazon fulfillment center entry on median household income and poverty rates within commuting zones. The dependent variables measure median household income in dollars and the poverty rate (the share of the population below the federal poverty line) at the commuting zone level. Data are from the American Community Survey (ACS) 1-Year estimates for 2014–2023, aggregated to the commuting zone level using person weights. The estimation sample consists of a balanced panel of 1,410 commuting zone-year observations from commuting zones that eventually receive Amazon facilities during the study period. Estimates are obtained from a staggered difference-in-differences specification following [Callaway and Sant’Anna \(2021\)](#), which compares treated commuting zones to not-yet-treated control units while incorporating commuting zone fixed effects and year fixed effects. Standard errors, reported in parentheses, are clustered at the commuting zone level using wild bootstrap procedures. Statistical significance levels: \*  $p < 0.10$ , \*\*  $p < 0.05$ , \*\*\*  $p < 0.01$ . Mean Dep. Var. reports the pre-treatment mean of the dependent variable in treated commuting zones. SD denotes the standard deviation of the dependent variable.

[Fajgelbaum and Gaubert \(2020\)](#) and [Rossi-Hansberg et al. \(2021\)](#) show these spillovers concentrate in high-skill, cognitive occupations that cluster geographically — the same pattern this section documents for Amazon.

A shift in establishment counts is suggestive but indirect. If the local economy is genuinely reorganizing toward skilled work, employment itself should move toward abstract, cognitive tasks rather than routine or manual ones. Figure 5 and Table 6 check this using the task measures of [Autor and Dorn \(2013\)](#). Abstract task share rises 4.15 percentage points after entry (significant at 5%); routine task share moves a smaller 1.49 points (significant at 10%); manual task share barely changes at all. The gap between these three estimates is the point — this is not a generic expansion of low-skill work, it’s a shift in the composition of what the local labor market does.

Figure 5: The Demand Side: Local Employment Shares by Task Type



*Notes:* This figure presents dynamic estimates of Amazon fulfillment center entry on local task composition using American Community Survey data for 2010–2023. The dependent variables measure percentage-point changes in commuting zone employment shares across three task categories: abstract, routine, and manual tasks. Task classifications follow [Autor and Dorn \(2013\)](#), with occupation-level task intensities weighted by local employment shares within each commuting zone and year. The estimation sample consists of a balanced panel of 170 commuting zones that eventually receive Amazon facilities during the study period. The vertical dashed line indicates the year of Amazon’s first fulfillment center opening in each commuting zone, with year  $t = -1$  serving as the omitted reference period. Estimates are obtained from a staggered difference-in-differences specification following [Callaway and Sant’Anna \(2021\)](#), which compares treated commuting zones to not-yet-treated control units while incorporating commuting zone fixed effects and year fixed effects. Each point represents an event-time average treatment effect on the treated (ATT), with vertical bars indicating pointwise 95% confidence intervals (point estimate  $\pm 1.96$  times the wild bootstrap standard error), clustered at the commuting zone level. Panel A shows that abstract task employment shares increase significantly in the post-treatment period, indicating growth in high-skill labor demand following Amazon entry. In contrast, routine and manual task shares exhibit no statistically significant changes.

Table 6: Effects of Amazon’s Entry on Local Task Composition

| Variable       | Abstract Task Share  | Routine Task Share  | Manual Task Share  |
|----------------|----------------------|---------------------|--------------------|
| Amazon Entry   | 4.1511**<br>(1.5423) | 1.4851*<br>(0.8643) | 0.2580<br>(0.3636) |
| Mean Dep. Var. | 188.31               | 251.35              | 78.48              |
| N              | 2,348                | 2,348               | 2,348              |

*Notes:* This table presents average treatment effects on the treated (ATT) for the impact of Amazon fulfillment center entry on local task composition. The dependent variables measure percentage-point changes in commuting zone employment shares across three task categories: abstract, routine, and manual tasks. Task classifications follow [Autor and Dorn \(2013\)](#), with occupation-level task intensities weighted by local employment shares within each commuting zone and year. The estimation sample consists of a balanced panel of 2,348 commuting zone-year observations from commuting zones that eventually receive Amazon facilities during the study period. Estimates are obtained from a staggered difference-in-differences specification following [Callaway and Sant’Anna \(2021\)](#), which compares treated commuting zones to not-yet-treated control units while incorporating commuting zone fixed effects and year fixed effects. Standard errors, reported in parentheses, are clustered at the commuting zone level using wild bootstrap procedures. Statistical significance levels: \*  $p < 0.10$ , \*\*  $p < 0.05$ , \*\*\*  $p < 0.01$ . Mean Dep. Var. reports the pre-treatment mean of the dependent variable in treated commuting zones.

The last piece connects this labor demand shift back to enrollment directly. If firms are hiring for cognitive, skilled work, that demand should show up in job postings before it shows up in degrees — and if students are responding to it, the two series should move together. Appendix Figure A3 plots the share of technology job postings against the share of the population holding a STEM master’s degree. The two track each other closely: postings begin rising within a year or two of entry, and STEM master’s attainment follows the same trajectory with a short lag. Alongside the establishment and task-share results, this is a fairly clean demand-pull story — Amazon’s presence raises the return to advanced, technical training, and the local population responds.

The mechanism evidence points to two margins operating side by side, as the conceptual framework’s second-margin discussion anticipated, rather than a single undifferentiated “upgrading” story. Rising income eases the constraint on staying in school and going to college, and a shift in local labor demand toward skilled, abstract work raises the payoff to a bachelor’s degree specifically — both operating through local residents’ own investment decisions, consistent with the timing of the bachelor’s-degree and enrollment results. The master’s-degree result, by contrast, is better read as evidence of the same demand-side transformation operating through a different margin: it draws already-credentialed workers in from elsewhere rather than generating new local investment, consistent with both its faster timing and the population evidence above.

## 5 Robustness Checks

This section examines the robustness of the main findings to alternative specifications and potential confounding factors. I focus on two primary concerns: the coincidence of Amazon’s rapid expansion with the COVID-19 pandemic, and the sensitivity of results to the choice of geographic unit of analysis. The county-level regression table is reported in the Online Appendix to conserve space; headline magnitudes are summarized here.

### 5.1 Excluding the COVID-19 Period

A key identification concern arises from the temporal overlap between Amazon’s most intensive expansion phase and the COVID-19 pandemic. The period 2020–2021 witnessed unprecedented disruptions to educational institutions and labor markets, including widespread shifts to remote learning and substantial enrollment volatility. Critically, this same period coincided with Amazon’s most dramatic network expansion, with facility openings during these two years accounting for approximately 43% of total network growth in the sample period (see Online Appendix Figure A6). This temporal overlap raises concerns that the baseline estimates may conflate pandemic-specific effects with the causal impact of Amazon entry. To address this concern, I re-estimate the baseline specifications on a sample that excludes observations from 2020 and 2021. If the documented educational upgrading reflects a genuine response to Amazon entry rather than pandemic-specific dynamics, the main results should persist in this restricted sample.

Appendix Table A2 presents estimates for educational attainment outcomes. The positive effects on higher education remain statistically significant and economically meaningful. The estimated effect on bachelor’s degree attainment is 0.61 percentage points, while the effect on master’s degree attainment is 0.47 percentage points. These estimates are comparable to, and in some cases slightly larger than, the baseline estimates reported in the main analysis.

Appendix Table A3 examines enrollment in four-year public colleges. The estimated effect on total enrollment is 0.58 percentage points, driven primarily by male enrollment, which increases by 0.37 percentage points. These patterns closely mirror those documented in the baseline analysis.

Together, these results provide strong evidence that the documented educational effects represent a robust consequence of Amazon fulfillment center entry rather than an artifact of pandemic-related disruptions.

### 5.2 County-Level Analysis

The baseline analysis is conducted at the commuting zone level to capture regional labor market effects and potential spillovers that extend beyond individual county boundaries. However, this

aggregation raises the question of whether the documented effects are observable at a more disaggregated geographic level. To address this concern, I re-estimate the baseline specifications using county-level variation.

The county-level analysis is more restrictive than the commuting zone approach, as it compares treated counties only to other untreated counties and does not capture spillovers within broader economic regions. Nevertheless, if the educational upgrading reflects genuine responses to Amazon entry, effects should be detectable at the county level, albeit potentially smaller in magnitude due to the exclusion of regional spillovers.

Online Appendix Table A4 presents county-level estimates for educational attainment. The results confirm the main findings: bachelor's degree attainment increases by 0.37 percentage points, and master's degree attainment increases by 0.22 percentage points. Effects on high school completion and some college attainment remain positive but statistically insignificant, consistent with the commuting zone-level results.

Two features of these county-level estimates warrant discussion. First, the statistical significance of the bachelor's and master's degree effects confirms that educational upgrading is observable at finer geographic scales and is not an artifact of commuting zone aggregation. Second, the magnitudes of the county-level estimates are smaller than their commuting zone counterparts. This attenuation is consistent with the presence of positive spillovers across county boundaries within commuting zones. The county-level estimate captures only the direct effect on host counties, while the commuting zone estimate incorporates both direct effects and spillovers throughout the regional labor market. This pattern supports the choice of the commuting zone as the appropriate unit of analysis for capturing the full economic impact of Amazon entry.

## 6 Conclusion

This paper estimates how Amazon's fulfillment center expansion affects local human capital formation. I link Amazon's facility-level rollout to county- and commuting-zone-level outcomes and exploit staggered entry timing across 170 commuting zones between 2010 and 2023. My estimates show that Amazon entry raises bachelor's degree attainment by 0.58 percentage points and master's degree attainment by 0.41 percentage points. I do not find evidence of the rising dropout rates that typically accompany local labor demand booms.

My results are consistent with a model in which income gains and skill-biased demand spillovers dominate the substitution effect emphasized in prior boom studies ([Black et al., 2005](#); [Charles et al., 2018](#); [Cascio and Narayan, 2022](#)). Median household income rises by roughly \$1,000 after entry, and the poverty rate falls, consistent with relaxed credit constraints on schooling. At the same time, the local economy shifts toward skilled work: information-sector establishments grow, employment

tilts toward abstract tasks, and technology job postings rise alongside STEM master's attainment. Four-year enrollment increases by 0.52 percentage points, with the gain concentrated among men; the racial breakdown skews similarly toward White students, while Black and Hispanic estimates are positive but imprecise. I complement these results with a population check that rules out large-scale in-migration as the primary driver: the 18–24 population shows no detectable change, though master's-degree holders rise 1.6 percent, pointing to some narrow in-migration at the top of the distribution. In total, my estimates indicate that Amazon's entry generates sizable, positive educational consequences for the communities that host its facilities, running counter to the pattern documented for resource and housing booms.

From a policy perspective, these results speak to the debate over whether attracting large e-commerce and logistics employers is worth the public subsidies typically required to do so. Of first-order concern in this debate is whether such firms help or harm human capital formation in the communities that host them. My results speak directly to this question in the context of e-commerce logistics. However, there are a couple of caveats to generalizing these findings. First, the effects I estimate are aggregate and commuting-zone-level. The population checks and the timing of the event-study results together suggest the bachelor's and enrollment results reflect local investment while the master's result is substantially a migration effect, but the decomposition rests on timing and an aggregate population count, not on data linking individuals to their own migration status and degree histories. Individual-level administrative data connecting Amazon's payroll records to workers' educational histories would sharpen this considerably. Second, my design identifies the effect of Amazon's expansion specifically, and it remains an open question whether other large logistics or technology employers would generate similar spillovers, or whether the pattern depends on features particular to Amazon's scale and operations. Third, FC siting is frequently accompanied by local rezoning, infrastructure investment, and other place-based policy changes intended to attract the facility in the first place; the reduced-form design here cannot fully separate Amazon's own causal contribution from these concurrent local development efforts, nor from broader regional growth in logistics-adjacent industries that may cluster around the same locations for reasons unrelated to Amazon specifically. Future work should also ask whether the size of these effects tracks the size of the subsidy package a locality offered, and why the gains go so disproportionately to men and white students.

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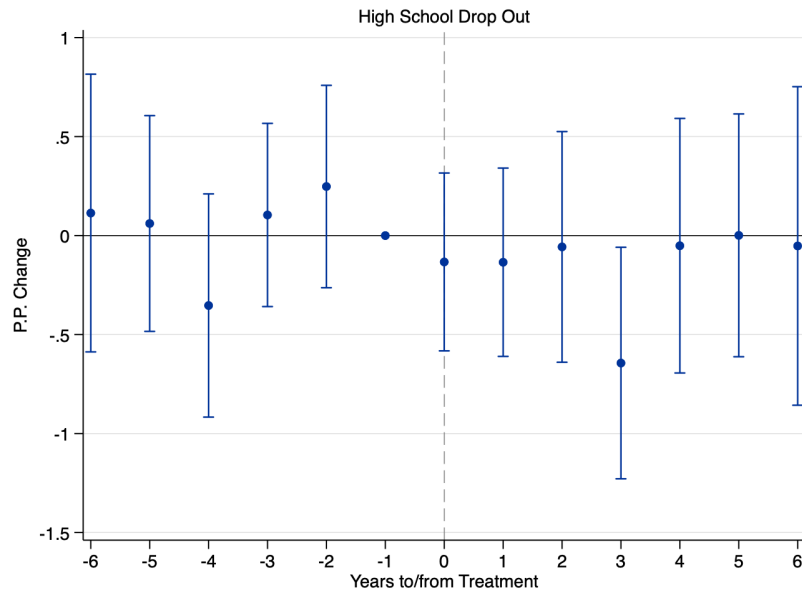
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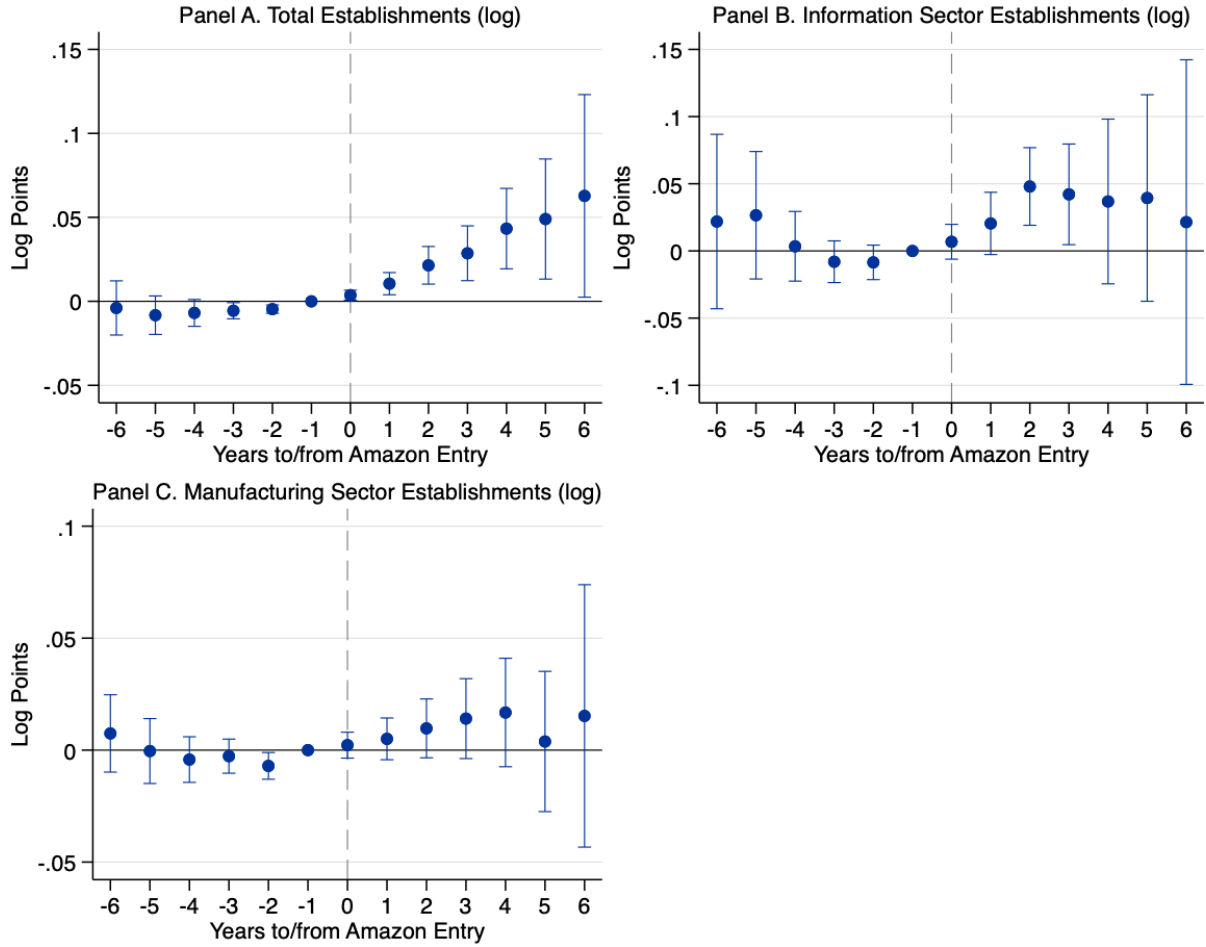
## Appendix

Figure A1: Event-Study Estimates: Effects of Amazon's Entry on High School Dropout Rates (Ages 16–19)



*Notes:* This figure presents dynamic estimates of Amazon fulfillment center entry on high school dropout rates among individuals aged 16–19 within commuting zones. The dependent variable measures the percentage-point share of individuals aged 16–19 without a high school diploma or equivalent, constructed by dividing the number of individuals without a diploma by the total population aged 16–19 in each commuting zone and year. Data are from the American Community Survey (ACS) 1-Year estimates for 2010–2023, aggregated to the commuting zone level using person weights. The vertical dashed line indicates the year of Amazon's first fulfillment center opening in each commuting zone, with year  $t = -1$  serving as the omitted reference period. Estimates are obtained from a staggered difference-in-differences specification following [Callaway and Sant'Anna \(2021\)](#), which compares treated commuting zones to not-yet-treated control units while incorporating commuting zone fixed effects and year fixed effects. Each point represents an event-time average treatment effect on the treated (ATT), with vertical bars indicating pointwise 95% confidence intervals (point estimate  $\pm 1.96$  times the wild bootstrap standard error), clustered at the commuting zone level.

Figure A2: The Demand-Side Spillover Channel: Changes in Establishments



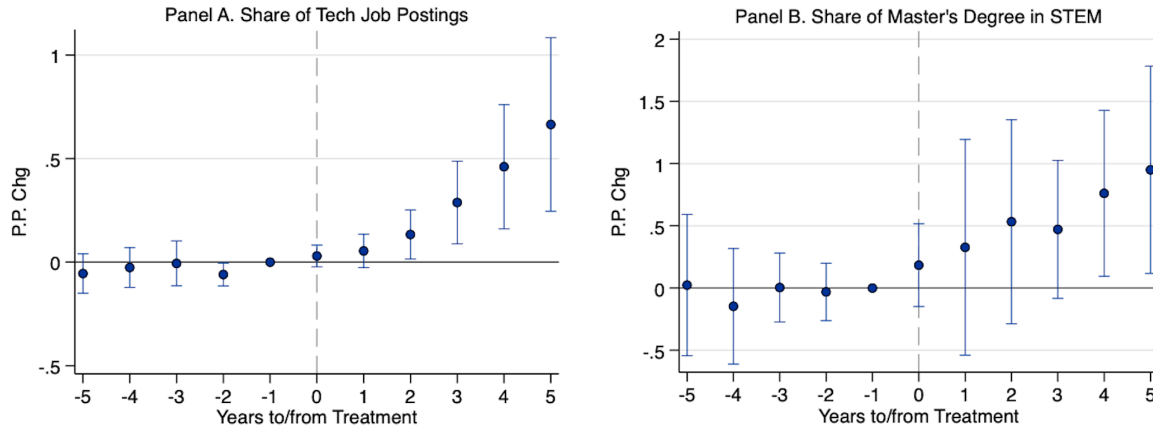
*Notes:* This figure presents dynamic estimates of Amazon fulfillment center entry on establishment counts across different sectors within commuting zones. The dependent variables measure log-point changes in the number of establishments within each sector: retail trade (NAICS 44-45), transportation and warehousing (NAICS 48-49), information (NAICS 51), professional and technical services (NAICS 54), and educational services (NAICS 61). Establishment counts are measured at the county level and aggregated to the commuting zone level by summing across all counties within each commuting zone. Data are from the County Business Patterns (CBP) for 2014–2023. The estimation sample consists of a balanced panel of 170 commuting zones that eventually receive Amazon facilities during the study period. The vertical dashed line indicates the year of Amazon’s first fulfillment center opening in each commuting zone, with year  $t = -1$  serving as the omitted reference period. Estimates are obtained from a staggered difference-in-differences specification following [Callaway and Sant’Anna \(2021\)](#). Each point represents an event-time average treatment effect on the treated (ATT), with vertical bars indicating pointwise 95% confidence intervals (point estimate  $\pm 1.96$  times the wild bootstrap standard error), clustered at the commuting zone level.

Table A1: Effects of Amazon's Entry on Sectoral Establishments

|                | Information (log) | Manufacturing (log) | Total Establishments (log) |
|----------------|-------------------|---------------------|----------------------------|
| Amazon Entry   | 0.0262*           | 0.0073              | 0.0188**                   |
|                | (0.0120)          | (0.0056)            | (0.0055)                   |
| Mean Dep. Var. | 5.17              | 6.08                | 9.33                       |
| N              | 1,410             | 1,410               | 1,410                      |

*Notes:* This table presents average treatment effects on the treated (ATT) for the impact of Amazon fulfillment center entry on establishment counts across selected sectors within commuting zones. Establishment counts are from County Business Patterns (CBP) for 2014–2023, aggregated to the commuting zone level. Estimates follow [Callaway and Sant'Anna \(2021\)](#). Standard errors, reported in parentheses, are clustered at the commuting zone level using wild bootstrap procedures. Statistical significance levels: \*  $p < 0.10$ , \*\*  $p < 0.05$ , \*\*\*  $p < 0.01$ . Mean Dep. Var. reports the pre-treatment mean of the dependent variable in treated commuting zones.

Figure A3: The Demand Side Channel: Tech Job Postings and STEM Master's Degrees



*Notes:* This figure presents dynamic estimates of Amazon fulfillment center entry on the share of technology job postings and STEM master's degrees awarded. Panel A examines technology job postings using data from Lightcast (formerly Burning Glass Technologies) for 2014–2023. Panel B analyzes STEM master's degrees using data from the Integrated Postsecondary Education Data System (IPEDS) for 2014–2023. The estimation sample consists of a balanced panel of 170 commuting zones that eventually receive Amazon facilities during the study period. Estimates are obtained from a staggered difference-in-differences specification following [Callaway and Sant'Anna \(2021\)](#).

Table A2: Robustness Check: Effects of Amazon’s Entry on Educational Attainment (Excluding 2020–2021)

|                | High School+ | Some College+ | Bachelor’s+ | Master’s+ |
|----------------|--------------|---------------|-------------|-----------|
| Amazon Entry   | 0.216*       | 0.318         | 0.612***    | 0.472***  |
|                | (0.113)      | (0.297)       | (0.165)     | (0.113)   |
| Mean Dep. Var. | 88.36        | 59.84         | 28.85       | 10.81     |
| SD             | 4.26         | 6.84          | 6.49        | 3.33      |
| N              | 1,934        | 1,934         | 1,934       | 1,934     |

*Notes:* Estimated on a sample excluding 2020–2021, during which approximately 43% of Amazon’s network expansion occurred, coinciding with COVID-19-related disruptions. Data from ACS 1-Year estimates (2010–2023, excluding 2020–2021). Estimates follow [Callaway and Sant’Anna \(2021\)](#). Standard errors, reported in parentheses, are clustered at the commuting zone level using wild bootstrap procedures. Statistical significance levels: \*  $p < 0.10$ , \*\*  $p < 0.05$ , \*\*\*  $p < 0.01$ .

Table A3: Robustness Check: Effects of Amazon’s Entry on Enrollment in 4-Year Public Colleges (Excluding 2020–2021)

|                | Total Enrollment | Male Enrollment | Female Enrollment |
|----------------|------------------|-----------------|-------------------|
| Amazon Entry   | 0.574**          | 0.373***        | 0.201             |
|                | (0.272)          | (0.133)         | (0.145)           |
| Mean Dep. Var. | 14.96            | 6.70            | 8.26              |
| SD             | 18.06            | 8.23            | 10.02             |
| N              | 1,944            | 1,944           | 1,944             |

*Notes:* Estimated on a sample excluding 2020–2021. Enrollment data from IPEDS (2010–2023, excluding 2020–2021); population denominators from ACS. Estimates follow [Callaway and Sant’Anna \(2021\)](#). Standard errors clustered at the commuting zone level using wild bootstrap procedures. Statistical significance levels: \*  $p < 0.10$ , \*\*  $p < 0.05$ , \*\*\*  $p < 0.01$ .

Table A4: Balance Tests for Educational Outcomes in First-Differences by Treatment Group

| Outcome (First-Differences)                | 2010–2013 (Cohort 1) |         |         | 2014–2016 (Cohort 2) |         |         | 2017–2019 (Cohort 3) |         |         |
|--------------------------------------------|----------------------|---------|---------|----------------------|---------|---------|----------------------|---------|---------|
|                                            | Treat                | Control | P-Value | Treat                | Control | P-Value | Treat                | Control | P-Value |
| <b>Panel A. Less than High School</b>      |                      |         |         |                      |         |         |                      |         |         |
| Combined                                   | -0.3854              | -0.2883 | 0.154   | -0.2817              | -0.1654 | 0.064   | -0.3805              | -0.2976 | 0.296   |
| Men                                        | -0.3973              | -0.2620 | 0.083   | -0.2506              | -0.0926 | 0.080   | -0.3966              | -0.3651 | 0.760   |
| Women                                      | -0.3725              | -0.3122 | 0.425   | -0.3130              | -0.2312 | 0.202   | -0.3630              | -0.2344 | 0.166   |
| <b>Panel B. High School Degree or More</b> |                      |         |         |                      |         |         |                      |         |         |
| Combined                                   | 0.3854               | 0.2883  | 0.154   | 0.2817               | 0.1654  | 0.064   | 0.3805               | 0.2976  | 0.296   |
| Men                                        | 0.3973               | 0.2620  | 0.083   | 0.2506               | 0.0926  | 0.080   | 0.3966               | 0.3651  | 0.760   |
| Women                                      | 0.3725               | 0.3122  | 0.425   | 0.3130               | 0.2312  | 0.202   | 0.3630               | 0.2344  | 0.166   |
| <b>Panel C. Some College or More</b>       |                      |         |         |                      |         |         |                      |         |         |
| Combined                                   | 0.5137               | 0.6322  | 0.233   | 0.4898               | 0.5009  | 0.918   | 0.4205               | 0.1940  | 0.109   |
| Men                                        | 0.3974               | 0.5488  | 0.110   | 0.3569               | 0.3058  | 0.707   | 0.3104               | 0.0414  | 0.141   |
| Women                                      | 0.6230               | 0.7110  | 0.497   | 0.6152               | 0.6798  | 0.579   | 0.5210               | 0.3428  | 0.263   |
| <b>Panel D. Bachelor's Degree or More</b>  |                      |         |         |                      |         |         |                      |         |         |
| Combined                                   | 0.3374               | 0.5950  | 0.000   | 0.4838               | 0.5436  | 0.433   | 0.5250               | 0.4907  | 0.755   |
| Men                                        | 0.1713               | 0.4124  | 0.002   | 0.3203               | 0.3784  | 0.562   | 0.3697               | 0.3718  | 0.988   |
| Women                                      | 0.4920               | 0.7642  | 0.004   | 0.6404               | 0.6942  | 0.520   | 0.6720               | 0.6001  | 0.612   |
| <b>Panel E. Master's Degree or More</b>    |                      |         |         |                      |         |         |                      |         |         |
| Combined                                   | 0.1718               | 0.3116  | 0.001   | 0.2250               | 0.2504  | 0.678   | 0.2233               | 0.2608  | 0.653   |
| Men                                        | 0.0777               | 0.1986  | 0.025   | 0.1465               | 0.1913  | 0.539   | 0.1517               | 0.0993  | 0.596   |
| Women                                      | 0.2596               | 0.4163  | 0.003   | 0.3004               | 0.3033  | 0.966   | 0.2919               | 0.4121  | 0.275   |

*Notes:* This table presents balance tests for pre-treatment trends in educational outcomes across treatment cohorts. Treatment cohorts are defined by the year of first Amazon fulfillment center entry: Cohort 1 comprises commuting zones first treated between 2014 and 2016, Cohort 2 between 2017 and 2019, and Cohort 3 between 2020 and 2023. For each cohort, balance is assessed using the corresponding pre-treatment period. P-values are from two-sample t-tests comparing mean first-differences between treated and control groups. Following [Marcus and Sant'Anna \(2021\)](#), these tests directly assess whether pre-treatment trends are balanced between treatment and control groups, a key requirement for the parallel trends assumption in staggered difference-in-differences designs. For the vast majority of outcomes and cohorts, we fail to reject the null hypothesis of balanced pre-treatment trends. The full balance-test appendix, including demographic covariates, is also reported in the Online Appendix.

Table A5: Balance Tests for Demographic Variables in First-Differences by Treatment Group

| Variable (First-Differences)     | 2010–2013 (Cohort 1) |          |         | 2014–2016 (Cohort 2) |         |         | 2017–2019 (Cohort 3) |         |         |
|----------------------------------|----------------------|----------|---------|----------------------|---------|---------|----------------------|---------|---------|
|                                  | Treat                | Control  | P-Value | Treat                | Control | P-Value | Treat                | Control | P-Value |
| Total Population                 | 6380.69              | 20278.86 | 0.002   | 5996.46              | 7384.71 | 0.454   | 5531.12              | 1824.94 | 0.009   |
| Male Population                  | 3278.07              | 10575.26 | 0.002   | 2902.92              | 3666.60 | 0.395   | 2742.14              | 884.99  | 0.009   |
| Female Population                | 3102.62              | 9703.60  | 0.003   | 3093.54              | 3718.11 | 0.519   | 2788.98              | 939.95  | 0.009   |
| Male Population, Age 18-24       | 342.34               | 682.43   | 0.242   | -186.47              | -218.93 | 0.806   | -226.81              | -110.27 | 0.426   |
| Male Population, Age 25-34       | 822.16               | 2551.91  | 0.012   | 719.82               | 937.79  | 0.348   | 640.68               | 228.23  | 0.012   |
| Female Population, Age 18-24     | 273.19               | 507.62   | 0.405   | -211.35              | -193.33 | 0.884   | -133.91              | -130.52 | 0.977   |
| Female Population, Age 25-34     | 631.89               | 2167.00  | 0.006   | 698.76               | 800.41  | 0.623   | 434.49               | 118.49  | 0.016   |
| White Population                 | 3244.67              | 9651.30  | 0.234   | 2513.02              | 5146.75 | 0.142   | 1956.19              | -406.65 | 0.146   |
| Black Population                 | 727.13               | 3043.00  | 0.019   | 1004.76              | 1189.92 | 0.655   | 752.23               | 370.62  | 0.301   |
| Hispanic Population              | 2596.14              | 6714.77  | 0.040   | 2991.57              | 3110.35 | 0.917   | 2894.73              | 768.49  | 0.012   |
| Female-to-Male Labor Force Ratio | -0.85                | -0.21    | 0.836   | -4.29                | -2.15   | 0.424   | -3.58                | -4.11   | 0.816   |
| Labor Force (16+)                | 4164.64              | 11581.08 | 0.017   | 3118.09              | 3613.54 | 0.626   | 4626.00              | 1389.14 | 0.002   |

*Notes:* This table presents balance tests for pre-treatment trends in demographic characteristics across treatment cohorts, defined analogously to Table A4. While some demographic variables show statistically significant differences in certain cohorts, reflecting heterogeneity in the size and composition of treated versus control commuting zones, the consistent patterns across cohorts support the validity of the identification strategy.